ence design which is licensed to SoC makers such as Qualcomm, Apple, Samsung and so on. Alongwith the processor designs, ARM also licenses the instruction sets to go along with the designs for third party SoC makers. They add on other components which will make up the SoC on top of the reference design and some like Qualcomm, even make changes to the processor micro-architechture. Apart from this, ARM licences its GPU reference designs as well.

## The main players

Like we said before the System-on-Chip involves many players who use ARM's designs and make their own SoCs. Qualcomm is the leader in this space and is the player to beat. It is to the mobile SoC space, what Intel is to the desktop CPU space. Qualcomm's Snapdragon series of SoCs were first seen in 2007 in phones such as HTC Tattoo back in 2007 and ever since the number of phone and tablet makers using Qualcomm Snapdragon SoCs has just increased exponentially. With innovations such as fabricating a 4G LTE chip right onto the SoC die, Qualcomm has had the first mover advantage. The current generation involves the Snapdragon 200, 400, 600 and 800 series.

NVIDIA Tegra is the next SoC which is quite popular. Thanks to its parent company which has a proven track record in the graphics industry, NVIDIA first implemented a Tegra SoC in Microsoft Zune back in 2009. Tegra 2 was showcased at CES 2010 and this was followed by the first quad-core plus one companion core SoC – the Tegra 3 which was launched in the second half of 2011. According to NVIDIA, the four + one core configuration involves a low power companion core which is put on the SoC which has the quad-core Cortex A9 cores. The low power companion core is used to handle low power tasks such as background processes when the phone is idle, tasks such as web browsing among other things. The higher powered quad-cores kick in when the task is resource intensive such as